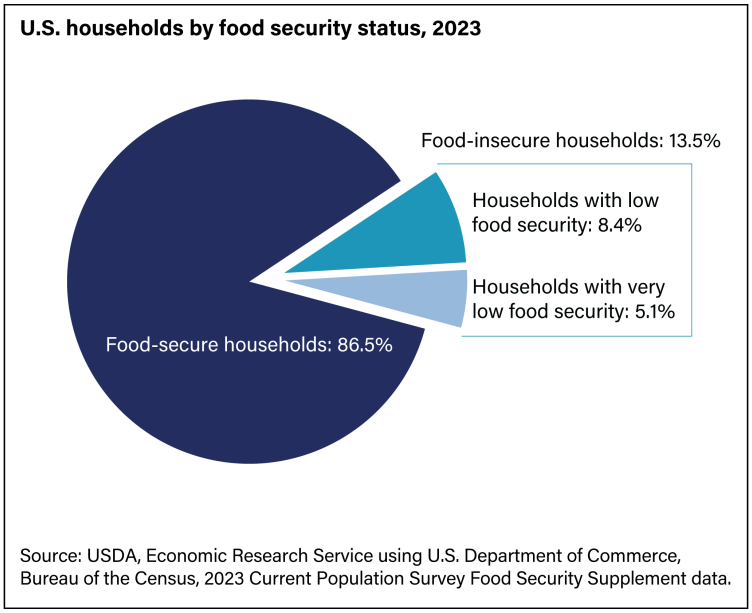
GitHub URL: https://github.com/dnyanadab/DATA-MINING-PROJECT-GROUP-13/tree/main
Website URL: https://when-food-exists-but-hunger-persists.base44.app

Introduction

- ## Research Topic & Significance:
  - The United States of America is the largest economy on earth with a GDP exceeding 31 trillion dollars, $90,000 per capita, split among nearly 350 million citizens. However, the abundance of wealth in the country does not translate to a universal supply of a basic need - food. Many of these 350 million people live in areas with limited access to food with related studies suggesting that as many as 18 million households were food insecure in 2023. This gap is driven primarily by socioeconomic factors, geographic differences, and infrastructure. Our primary goal for this project is to understand the impact of these three factors on relative food access in the richest nation in the world as well as its effects. We aim to visualize the often underreported reality of poverty and food accessibility in the US with the hopes of representing those who suffer the most from systemic problems in food access. Although the overarching goal of the project is to help those who suffer from poor food access, the main beneficiary of the research are the individuals who can make a difference through insightful and motivating reporting. We believe that through factual and applicable reporting on this topic, we can encourage individuals to take action and help address the problem of food insecurity through individual action and policy shifts.



- ## Stakeholders (Who is Affected?):
    - The primary focus of the project are those who currently face low access to food through infrastructure and/or socioeconomic impacts. These individuals, 18 million households in 2023, are those who suffer the most from this issue and the group who we aim to report the impacts on. Secondary stakeholders include the State, Federal, and Local government who provide a legislative framework in which food insecurity is indirectly influenced. In addition corporate entities, such as Wal-Mart and Dollar General, are often the groups who directly influence food insecurity through pricing and strategic location choice. Foodbanks are another stakeholder who work to fill the gaps where food insecurity is a problem. Organizations such as Feeding America operate many of these food banks. Food insecurity is a very real problem that often is underreported or accepted as an unfortunate reality. Our aim is to report on the topic in a way that impacts the actions of the end consumers of this information. As the final stakeholder of the project is the end consumer, the report must be designed in a way that is both generally understandable, impactful, and transparent. While we work to understand the gaps in food access, we also need to understand how to effectively empower the end user to promote change.

- ## Existing Solutions & Gaps:
    - Although food inaccessibility is a longstanding issue often related to deep rooted socioeconomic problems - there are organizations and policies in place to curb the impact. Federal programs such as SNAP provide assistance to low income individuals to buy meals from their grocery store. SNAP has been shown to not only reduce the impact of food insecurity, but also lift individuals out of poverty (SNAP lifts families out of poverty, bolsters local economies, expert says). Secondly, there are locally run food banks and soup kitchens who distribute food in areas with low food access. These are often run by local level organizations, but can often be coordinated with larger non-profits like Feeding America. Non-Profits with this mission provide for over 53 million Americans facing high food inaccessibility (53 Million People Visited Food Banks | Feeding America). While these organizations do a lot to increase food access, there are still issues that remain with food distribution. This includes economic factors like cost of living or the lack of infrastructure in a county to provide adequate food access. The regions that often face the highest food insecurity are regions in the US in which access to SNAP eligible markets can be very limited, especially to individuals without access to reliable transportation (Neighborhood Dynamics in SNAP Participation and Food Access | Chicago Council on Global Affairs).

- # Blueprint for Your Project

The project team plans to explore these disconnects by integrating multiple large-scale datasets to create a comprehensive analytical table at the county level. The high-level plan involves developing two primary composite metrics: a Food Infrastructure Index (FII), which averages the density of grocery stores, supercenters, and SNAP retailers, and a Low-Access Indicator (LAI), which standardizes USDA percentages of underserved populations. By calculating an "Access Mismatch Score," the team will identify counties in the top quartile for both infrastructure and access constraints, effectively highlighting where the system is failing most prominently. The analysis will also incorporate socioeconomic controls

such as vehicle access and median household income to ensure the findings distinguish between physical access issues and general poverty. Finally, the team will examine the geographical clustering of these mismatches and the specific role that predatory grocery chains play in these regions. The ultimate goal is to produce a ranked list of mismatched counties and visualizations that can inform more effective, data-driven food policy.

The overarching aim is to determine counties where food infrastructure is apparently sufficient but where access is still limited for individuals residing there. For this purpose, the team is to create two combined figures:

- Food Infrastructure Index (FII) – a standardized composite measure derived from the per-capita density of grocery stores, supercenters, and SNAP-authorized retailers (from the USDA Food Environment Atlas).
- Low Access Indicator (LAI) – a standardized composite constructed using USDA Food Access Research Atlas variables to represent low-income and low-access proportions in rural and urban settings.

Both will be standardized using z-score normalization to facilitate inter-county comparison. Subsequently, the Access Mismatch Score will be calculated and defined as the difference between standardized infrastructure and access measures, i.e., (FII - LAI). Counties ranked in the top quartile on both measures will be classified as “high-mismatch” areas where food systems are perceived to be robust on paper but where underlying access challenges are severe.

Control variables will be drawn from the American Community Survey to differentiate between physical access constraints and more general socioeconomic disadvantage, including median household income and percentage of households without vehicle access. These controls will help assess whether mismatches persist even after accounting for income and transportation limitations.

The project will then conduct spatial exploratory analysis, regional clustering, and choropleth mapping to gauge geographic patterns of mismatch. Using the Dollar General store location data, the project will assess the concentration of predatory or low-nutrition food retailers to test whether such chains are disproportionately present in high-mismatch counties.

Final outputs will include:

- A ranked list of counties with the most severe access–infrastructure disconnects
- Summary tables for diagnosis
- Geographic visualizations of clusters of systemic food access failure

These outputs are meant to help drive policymakers, public health agencies, and nutritional assistance organizations towards intervening in places where the food systems fail despite an apparent supply.

Datasets used:

- **USDA Food Access Research Atlas (2019)**: Used to measure structural food access constraints, including the count and percentage of low-income and low-access populations in both urban and rural settings.
- **USDA Food Environment Atlas (2020)**: Used to measure physical and institutional food infrastructure, specifically looking at the density of grocery stores, supercenters, and SNAP-authorized retailers.
- **American Community Survey (ACS) 2015-2019**: Provides critical socioeconomic and transportation context, such as median household income and the percentage of households without vehicle access.
- **Dollar General Locations Database**: A specialized dataset from Pareto Software used to track the concentration of these stores in relation to county-level food access and income

Research Questions:

Primary Research Question(RQ1)

Where do structural mismatches between food infrastructure availability and food access persist in counties across the U.S., characterized by standardized composite indexes of food supply (Food Infrastructure Index, FII), and food access constraint indexes (Low Access Indicator, LAI)?

Secondary Research Questions:

RQ2:
How much do mismatches in food access infrastructure matter after socioeconomic and transportation variables are controlled for, such as household income and access to automobiles?

RQ3:
Is the geographical distribution of counties with high access-infrastructure mismatch clustered, suggesting systemic rather than local failures?

RQ4:
What is the difference in the density of low-cost, low-nutrition food retailers (for example, Dollar General) between high-mismatch counties and other counties?

RQ5:
What food infrastructure components (grocery store density, supercenter density, SNAP-authorized retailer density) exhibit the weakest relationship to the conditions being

mitigated by changes in access constraints? RQ Are there distinctions between urban and rural counties in the degree to which food access and infrastructures mismatch?

RQ6:
Are there any findings suggesting that high-mismatch counties have compounding structural disadvantages, such as greater poverty levels or a lack of transportation, as opposed to low-mismatch counties?

RQ7:
How sensitive are mismatch classifications to alternative operational definitions of “high mismatch” (e.g., top decile vs. top quartile thresholds)?

Methodology Overview:

This research adopts a county-level multi-source data integration and exploratory modeling strategy. Data from all sources will be processed and integrated, and county Federal Information Processing Standards (FIPS) codes will be the main geographic identifier. Variables will be normalized according to each data source. Two different composite indices will be computed. These will be expressed as: Food Infrastructure Index (FII), which equals the average of z-normalized per capita densities of grocery stores, supercenters, and SNAP-authorized stores; and Low Access Indicator (LAI), which equals the average of z-normalized USDA low-income and low-access proxy variables for urban and rural populations.

There will be an Access Mismatch Score derived from the difference between FII and LAI. Counties are considered “high mismatch” if they have high values in both FII and LAI distributions, indicating that these are at the upper quartile range. Socioeconomic control factors from the American Community Survey, such as median household income and proportion of households without vehicle access, will be considered to see if the effect exists after adjusting for basic poverty levels.

This will be further supported by exploratory data analysis: distributional analysis, correlation testing, and geographic visualization that seek to identify spatial clustering of mismatched counties. Finally, Dollar General store location data will be spatially aggregated to the county level and compared against mismatch classifications to evaluate whether low-nutrition food retailers are disproportionately concentrated in structurally underserved regions.